Same Facts, Different Interfaces: Evaluating FAQ, Retrieval Chatbot, and AI Chatbot for Student Services

ABSTRACT

Universities rely heavily on online tools to answer common questions about admissions, tuition, registration, transcripts, housing, and campus services. However, it remains unclear how different interfaces compare in helping users access this information with less effort and greater trust. This study compared a frequently asked questions (FAQ) webpage, a retrieval chatbot, and an AI chatbot using a shared knowledge base developed for a hypothetical university. Forty participants completed the same student service information-seeking tasks across the three interfaces and a user-experience survey. The retrieval chatbot received the highest mean ease-of-use rating (4.03 out of 5) and the lowest reported-effort rating (2.75). The AI chatbot received the highest ratings for perceived speed (4.40), clarity (4.50), trust (4.50), and satisfaction (4.40), although some participants found its responses too wordy. The chatbots received more favorable ratings than the FAQ webpage across most measures, but no interface received the highest ratings across all measures. The findings highlight differences in perceived usability, effort, speed, and trust across the three interfaces.

1.0 INTRODUCTION

Universities use websites and other digital resources to provide information about admissions, registration, fees, and campus services (Peyton et al., 2025). Frequently asked questions (FAQ) webpages organize answers that users browse or search (Han & Lee, 2022). Chatbots offer a conversational alternative for accessing this information (Oliveira & Matos, 2023; Peyton et al., 2025). Retrieval chatbots return stored answers, while AI-powered chatbots can generate explanations based on available information. However, making information available does not necessarily make the relevant answer easy to find or understand.

To understand which interface best meets users' needs, comparisons must consider both the information provided and how users access it. Previous studies have examined different aspects of this question. In a course setting, Han and Lee (2022) found lower intention to use an FAQ chatbot and greater perceived barriers than for an FAQ webpage, identifying unfamiliarity and privacy concerns as possible explanations. In academic services, Suherwin et al. (2026) found that a retrieval-augmented generation chatbot produced more accurate answers than a stored-answer FAQ chatbot but responded more slowly. These findings suggest that offering a conversational interface or more accurate answers does not necessarily provide a better experience in every aspect. Comparing interfaces therefore requires considering both system performance and user experience. Research in other information-seeking settings has examined mental effort and trust. For instance, Stadler et al. (2024) found lower cognitive load among students using ChatGPT than

Google Search in a scientific inquiry task, while Li and Aral (2025) found that users' trust differed when identical information was presented as AI-generated or traditional search results.

Altogether, these studies support comparing interfaces through measures of accuracy, speed, mental effort, ease of use, and trust. However, the studies reviewed examined different systems and tasks, making it unclear how an FAQ webpage, retrieval chatbot, and AI chatbot compare across these measures when providing the same student-service information. This study therefore aims to compare the three interfaces using a shared knowledge base to identify which approach best helps users obtain correct answers quickly, with less mental effort, greater ease of use, and greater trust.

The objectives are to:

I. Develop an FAQ webpage, retrieval chatbot, and AI chatbot using a shared student-service knowledge base.
II. Compare participants' ratings of perceived speed, mental effort, ease of use, clarity, trust, accuracy, and satisfaction across the three interfaces.
III. Identify users' preferred interface and examine the reasons for their preferences.

2.0 METHODOLOGY

2.1 Study design

The study used an online comparative design in which participants interacted with an FAQ webpage, a retrieval chatbot, and an AI chatbot. Each participant used all three interfaces and evaluated their experience after completing student-service information tasks. The interfaces used a shared knowledge base, allowing the study to compare different ways of accessing the same underlying information.

2.2 Knowledge base and interface development

A student-service knowledge base was developed for a hypothetical university. It contained information about services such as registration, tuition payments, transcripts, housing, and academic advising. This provided a common information source for the three systems.

The FAQ webpage presented stored questions and answers that users could browse or search. The retrieval chatbot allowed users to enter questions in a chat interface and receive matching stored answers. The AI chatbot generated conversational responses based on the available information and encouraged users to ask follow-up questions. A common landing page provided access to all three interfaces.

2.3 Participants

The final sample consisted of 40 respondents. The questionnaire collected participants' highest level of education, current university-student status, and expected graduation year where applicable. Participants were instructed to assume the role of a student seeking information from the hypothetical university.

2.4 Tasks and procedure

Google Forms was used to present the study instructions, link participants to each interface, collect task answers, and record their evaluations. The form presented the interfaces in the following order: FAQ webpage, retrieval chatbot, and AI chatbot.

For each interface, participants read the task questions, opened the corresponding system, searched for answers, and returned to the form to enter their responses. They were instructed to provide brief answers, although copying and pasting from the system was also permitted. They were also asked to answer as accurately and quickly as possible without using external sources such as Google. Each interface section contained five student-service questions. Across the sections, these covered registration and course withdrawal periods, first-year housing requirements, transcript requests, accepted payment methods, and academic advising. After completing the tasks for each interface, participants rated their experience. Once they had used all three systems, they completed a final comparison section.

2.5 User-experience measures

Participants rated six statements on a five-point scale numbered from 1 (strongly disagree) to 5 (strongly agree):

1. The system was easy to use.
2. I was able to find information quickly.
3. The information provided was clear and understandable.
4. The system required a lot of effort to use.
5. I trust the information provided by this system.
6. I am satisfied with this system.

The final comparison asked participants to select the interface they found easiest to use, trusted most, considered fastest for finding answers, and preferred overall.

3.0 Results

3.1 Demographic Characteristics

Forty individuals participated in the survey (figure 2). Their highest completed education was high school for 31 participants (77.5%), an undergraduate degree for eight (20.0%), and a master's degree for one (2.5%). At the time of the study, 36 participants (90.0%) were enrolled in an educational program, while four (10.0%) were not.

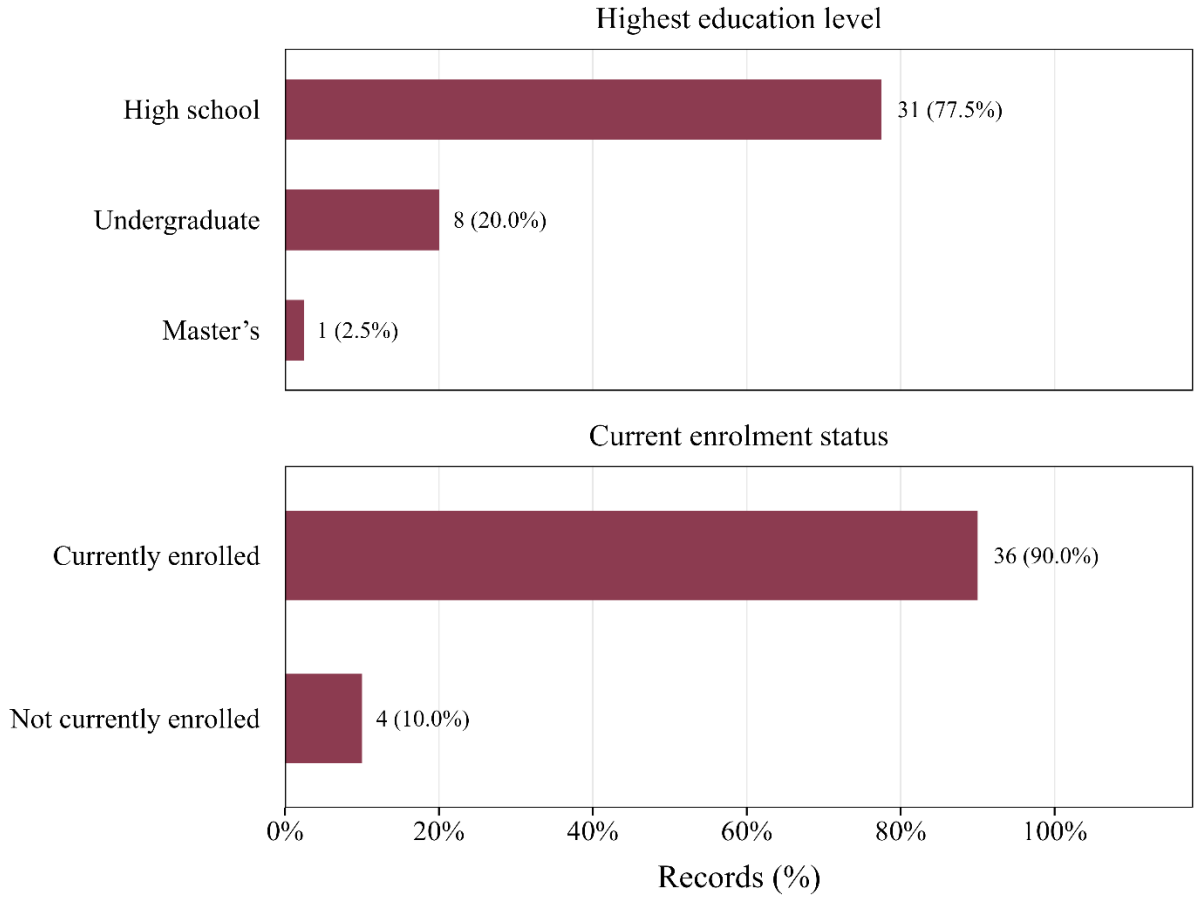


Figure 2: Distribution of the 40 participants by highest completed education and enrolment status at the time of the study. Labels show counts and their corresponding percentages.

Figure 3 presents the mean ratings for the three interfaces on a five-point scale. The retrieval chatbot received the highest ease-of-use rating (4.03), followed by the AI chatbot (4.00) and FAQ webpage (3.25). The AI chatbot received the highest ratings for perceived speed (4.40), clarity (4.50), trust (4.50), and satisfaction (4.40). The retrieval chatbot received the lowest effort rating (2.75), followed by the AI chatbot (3.00) and FAQ webpage (3.18).

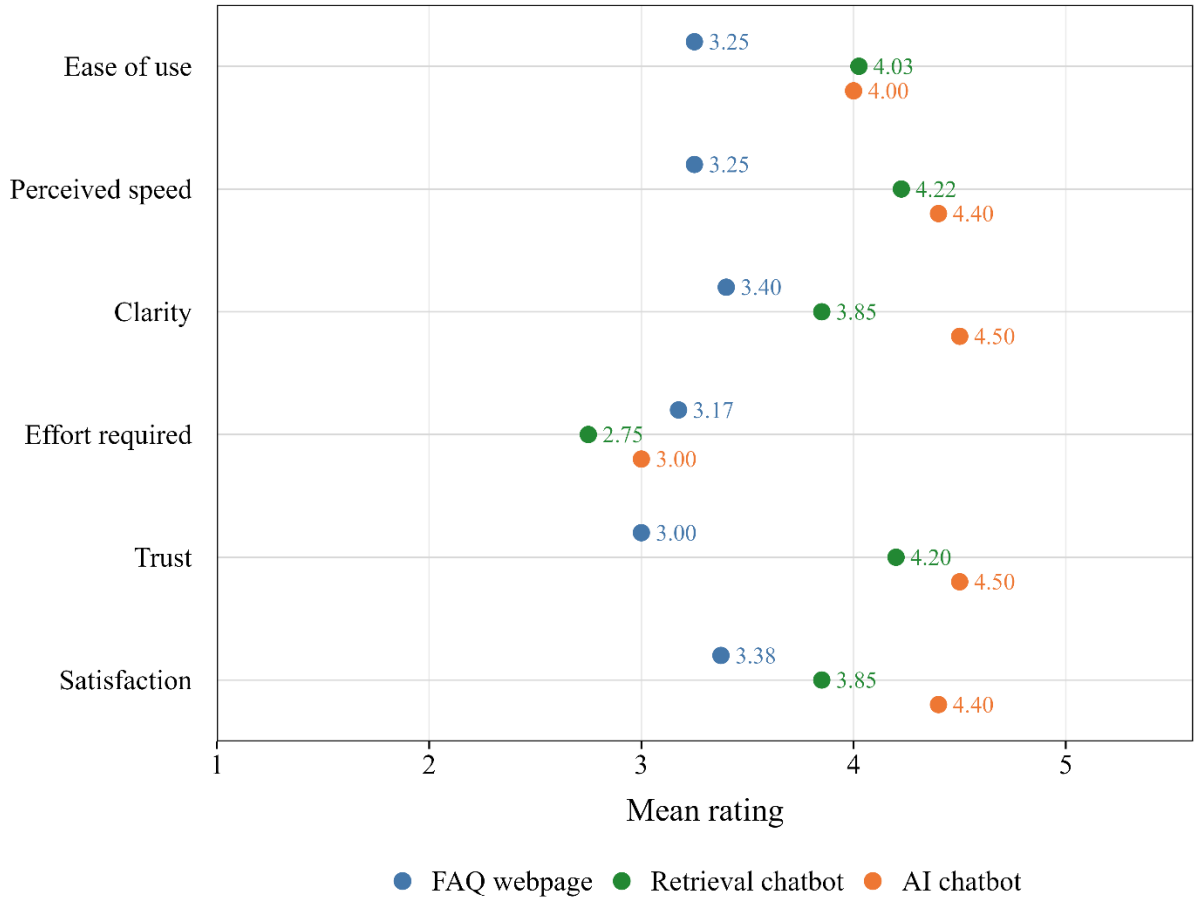


Figure 3: Mean ratings for the three interfaces on a five-point scale. Ratings ranged from 1 (strongly disagree) to 5 (strongly agree).

3.2 Direct Comparison of the Interfaces

The AI chatbot was selected as easiest to use and fastest for finding answers by 47.5% of participants, followed by the retrieval chatbot (42.5%) and FAQ webpage (10.0%). The AI chatbot also received the most selections for trust (47.5%), followed by the FAQ webpage (40.0%) and retrieval chatbot (12.5%).

Among participants who selected one of the three interfaces, 52.8% preferred the AI chatbot, 36.1% preferred the retrieval chatbot, and 11.1% preferred the FAQ webpage (Figure 4).

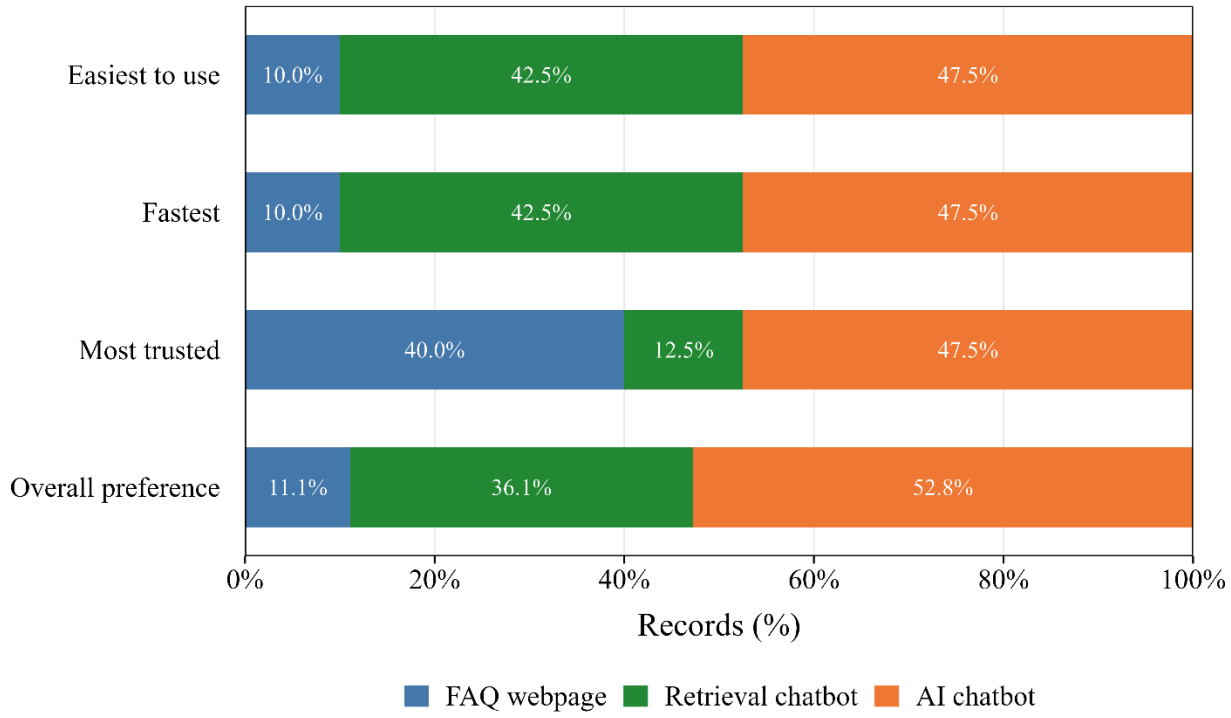


Figure 4: Respondents' selections for ease of use, speed, trust, and overall preference.

3.3 Comparisons by Highest Completed Education

Among participants whose highest completed education was high school, the AI chatbot received 61.3% of selections for easiest to use, fastest, most trusted, and overall preference. The retrieval chatbot received 25.8% of selections for ease, speed, and preference, while the FAQ webpage received 12.9%. Trust selections were divided between the AI chatbot (61.3%) and FAQ webpage (38.7%).

In the undergraduate and master's groups, all selections for easiest and fastest favored the retrieval chatbot. Undergraduate trust selections were divided equally between the FAQ webpage and retrieval chatbot, while the master's participant selected retrieval. All named overall-preference selections in both groups favored retrieval (Figure 5).

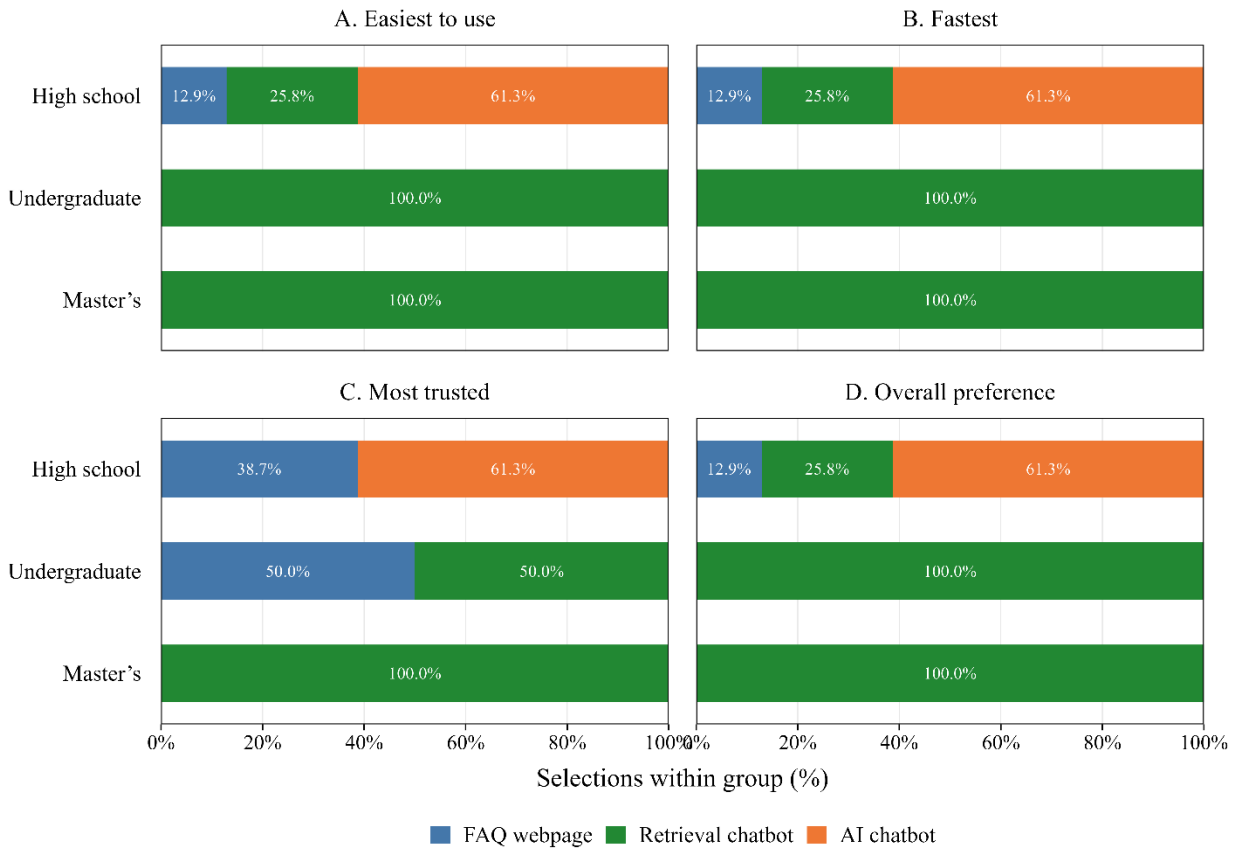


Figure 5. Interface selections by highest completed education for ease of use, perceived speed, trust, and overall preference.

Mean ratings also varied across education groups. In the high-school group, the AI chatbot received the highest ratings for ease, perceived speed, clarity, trust, and satisfaction, while retrieval received the lowest effort rating. In the undergraduate group, retrieval received the highest ease and perceived-speed ratings; clarity, trust, and satisfaction ratings were equal across interfaces. The master's participant rated both chatbots higher than the FAQ webpage for ease, perceived speed, and satisfaction, and reported less effort when using them (Figure 6).

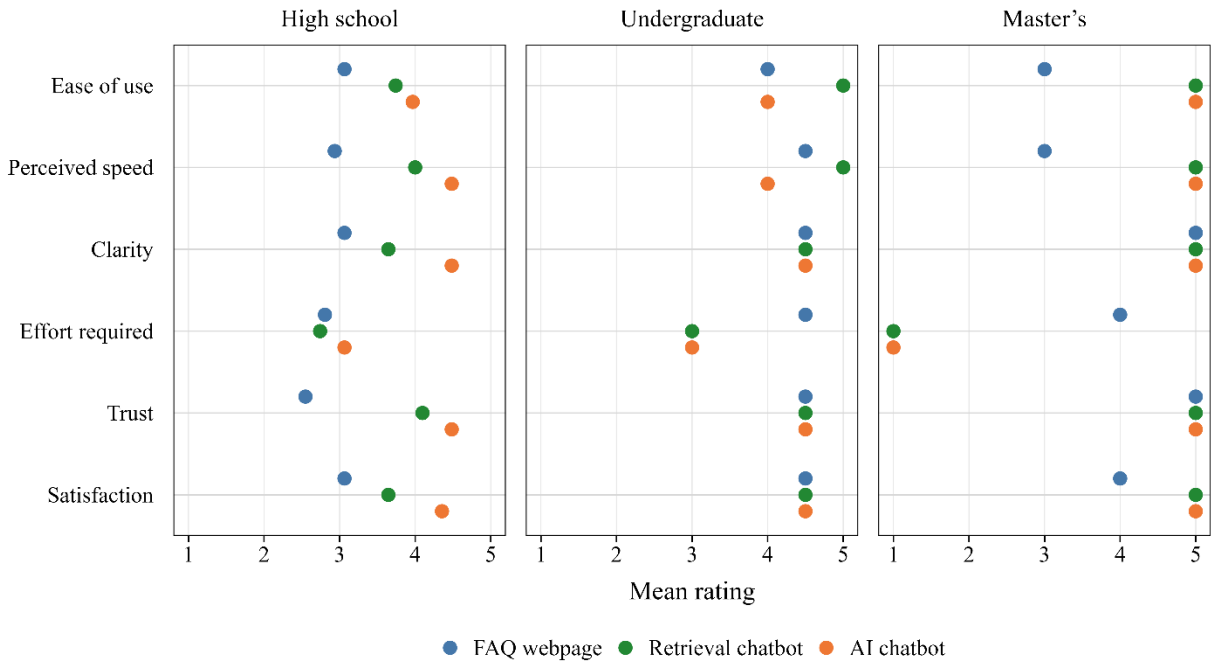


Figure 6. Mean interface ratings by highest completed education on a five-point scale. Ratings ranged from 1 (strongly disagree) to 5 (strongly agree)

3.4 Task Responses

Participants' submitted answers varied across questions and interfaces. Some FAQ entries stated that no matching answer was found or contained "Nothing." Others provided information unrelated to the question, such as transcript information in response to a registration question.

Retrieval-chatbot entries frequently contained brief stored answers. AI-chatbot entries ranged from short factual answers to longer conversational responses with introductory phrases, analogies, and offers of further assistance.

These observations describe participants' submissions rather than independently verified system outputs. Without scored answers or recorded task-completion times, they do not establish which interface was most accurate or fastest.

4.0 DISCUSSION

The two chatbots received more favorable ratings than the FAQ webpage across most measures, but their strengths differed. The retrieval chatbot received the highest ease-of-use rating and required the least reported effort, while the AI chatbot received the highest ratings for perceived speed, clarity, trust, and satisfaction. These findings differ from Han and Lee (2022), who found lower intention to use an FAQ chatbot and greater perceived barriers than an FAQ

webpage in a course setting. The findings suggest that the benefits of conversational access depend on the system and setting rather than the interface type alone.

The retrieval chatbot's brief stored answers may have contributed to its lower effort rating, while the AI chatbot's fuller explanations may have supported clarity and satisfaction. However, some participants found the AI responses too wordy, and response length was not systematically measured. Ghosh and Shah (2025) found that explicit system clarifications reduced the overall experience in spoken conversational search, suggesting that additional explanation does not always benefit users. Lower effort also does not necessarily indicate better task outcomes: Stadler et al. (2024) found lower cognitive load with ChatGPT than Google Search but weaker scientific justifications. These studies support evaluating whether explanations help users complete their tasks, rather than assuming that either shorter or longer responses are better.

Similarly, higher trust ratings indicate greater confidence in the information, but do not establish that the answers were more accurate. Li and Aral (2025) found that reference links increased trust in generative search even when the references were incorrect. Si et al. (2024) found that LLM explanations supported faster fact-checking, but participants relied too heavily on incorrect explanations. These findings support assessing trust and accuracy separately. The present study cannot establish whether the most trusted interface was the most accurate because numerical accuracy scores were unavailable.

The demographic comparisons suggest that interface preferences were not uniform across education groups. However, the sample was dominated by high-school graduates and currently enrolled participants, and only one participant held a master's degree. The small, uneven groups limit broader interpretation; the master's-degree findings represent one person's experience. These comparisons should therefore be treated as descriptive rather than evidence that education or enrollment determines interface preference.

Although the knowledge base reflected typical university services, the interfaces were not evaluated in an actual university setting. Task-completion times were not recorded, so speed findings reflect participants' perceptions.

4.0 CONCLUSION AND FUTURE WORK

This study compared an FAQ webpage, retrieval chatbot, and AI chatbot using a shared student-service knowledge base. The chatbots received more favorable ratings across most measures. The retrieval chatbot was rated easiest to use and required the least reported effort, while the AI chatbot received the highest ratings for perceived speed, clarity, trust, and satisfaction. No interface received the highest ratings across all measures, highlighting the need to consider several aspects of user experience when designing student-information services. These findings mostly describe participants' perception.

Future work should record task-completion times, score answers against the knowledge base, and evaluate the interfaces with actual university information. A larger, more balanced sample would strengthen comparisons across user groups. Validated cognitive-load measures, combined with task performance and behavioral measures such as navigation steps and repeated queries, would provide a fuller assessment of mental effort than self-reported ratings alone.

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